The Electron Mass from Phase Closure: Self-Consistent Torus Knots in the Superfluid Vacuum

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Abstract

We determine the electron mass from three self-consistency conditions on torus knot vortices in a superfluid vacuum. The superfluid condensate creates a graded-index refractive profile $n(r) = \sqrt{1 + \xi^2/r^2}$ around any depletion, where $\xi = \lambda\!\!\!^{-}_C = \hbar/(m_e c)$ is the healing length. A photon propagating on a $(2, 1)$ torus knot in this medium accumulates optical phase $\Phi = 2\pi p\sqrt{\beta^2 + 1}$, where $\beta = R/\xi$ is the orbit radius in units of the healing length. Phase closure ($\Phi = 2\pi m$, integer m) at the first physical resonance $m = 3$ gives $\beta = \sqrt{5}/2$, encoding the Pythagorean identity $3^2 = (\sqrt{5})^2 + 2^2$. Requiring simultaneous closure for the $(1, 4)$ proton knot at $m = 2$ on the same orbit determines the torus aspect ratio $\kappa = R/r = 12/\sqrt{7}$, with $\kappa^2 = 144/7 = (q_p m_e)^2/((m_p p_e)^2 - (p_p m_e)^2)$ expressed entirely in terms of topological winding numbers and mode integers. Finally, the superfluid vortex ring energy $E = (\rho_0 \kappa_v^2 R/2)[\ln(8R/\xi) - C]$ with $\rho_0 = 1.69 \times 10^5$ kg/m^3 and $C = 0$ yields $m_e = 9.19 \times 10^{-31}$ kg, matching experiment to 0.85%. The electron mass is thus determined by three inputs: the vacuum condensate density, the $(2, 1)$ torus knot topology, and the requirement that electron and proton resonate on the same orbit.

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1 Introduction

The electron mass $m_e = 9.109 \times 10^{-31}$ kg is one of the fundamental constants of nature. In the Standard Model it enters as a free parameter—the Yukawa coupling to the Higgs field—with no theoretical prediction of its value. Here we show that in the Null Worldtube (NWT) framework, m_e is *determined* from three self-consistency conditions on torus knot vortices in a superfluid vacuum.

The NWT framework models fundamental particles as quantized vortex rings in a superfluid vacuum condensate [1, 2]. The electron corresponds to a $(2, 1)$ torus knot—a vortex line that wraps twice around the major axis and once through the tube. Previous work demonstrated that, given m_e and α as inputs, the framework reproduces the masses of all Standard Model fermions to percent-level accuracy [3, 4]. The present paper eliminates m_e as a free parameter.

The derivation proceeds in three steps:

1. **Phase closure** (Section 3): A photon traveling along the $(2, 1)$ knot in the superfluid refractive profile must form a standing wave. This fixes the orbit radius $R = (\sqrt{5}/2)\,\xi$ where $\xi = \lambda\!\!\!^{-}_C$ is the healing length.

2. **Dual resonance** (Section 4): Requiring the $(1, 4)$ proton knot to simultaneously resonate at the same orbit fixes the torus aspect ratio $\kappa = 12/\sqrt{7}$.

3. **Vortex energy** (Section 5): The superfluid vortex ring energy, with core cutoff at ξ, gives m_e in terms of the condensate density ρ_0.

2 The Superfluid Vacuum

We model the quantum vacuum as a Bose–Einstein condensate described by the Gross–Pitaevskii equation:

$$i\hbar\,\partial_t\psi = -\frac{\hbar^2}{2m^*}\nabla^2\psi + g|\psi|^2\psi \tag{1}$$

where $m^* = m_e/\sqrt{2}$ is the effective boson mass and g is the self-interaction coupling. The equilibrium condensate has uniform density $n_0 = \rho_0/m^*$ and healing length

$$\xi = \frac{\hbar}{\sqrt{2m^*\,g\,n_0}} = \frac{\hbar}{m_e c} = \lambda\!\!\!^{-}_C = 386.3 \text{ fm} \tag{2}$$

identifying the healing length with the reduced Compton wavelength. The identification $\xi = \lambda\!\!\!^{-}_C$ requires $g\,n_0 = m^*c^2$, fixing the sound speed $c_s = \sqrt{g\,n_0/m^*} = c$: phonon excitations of the vacuum condensate propagate at the speed of light, as required by Lorentz invariance at leading order. The condensate density $\rho_0 = n_0 m^*$ then enters as the single dimensionful parameter of

the vacuum model. Its value is determined self-consistently: $g\,n_0 = m^* c^2$ fixes the coupling in terms of the density, and the healing length condition (2) then gives

$$\rho_0 = \frac{m_e^3\, c^2}{4\sqrt{2}\,\hbar\, g} \tag{3}$$

The coupling g is not independently measured for the vacuum condensate, so ρ_0 (equivalently g) remains the single free parameter. In prior work [4], matching the vortex ring energy to $m_e c^2$ with the tube radius $r = R/\pi^2$ gave $\rho_0 = 1.69 \times 10^5$ kg/m^3 ($g = 2.2 \times 10^{-49}$ J m^3). The present derivation uses this ρ_0 as input and recovers m_e to 0.85%; the residual error corresponds to a 3.4% shift in ρ_0 (equivalently in g), well within the uncertainty of the original calibration.

2.1 Refractive index from the vortex density profile

A quantized vortex of winding number ν in the GPE has the asymptotic density profile $|\psi|^2 = n_0\, f^2(r)$, where $f(r) \to (r/\xi)^\nu$ as $r \to 0$ and $f \to 1$ as $r \to \infty$. The standard Padé interpolant matching both limits is [7]

$$f(r) = \frac{r}{\sqrt{r^2 + \nu\,\xi^2}} \tag{4}$$

which agrees with numerical GPE solutions to $\lesssim 5\%$ for all r [8].

The local sound speed in the condensate scales with the density as $c_s(r) = c\, f(r)$. Phonon-like excitations propagating through the depletion zone therefore see the phase velocity $v_{\rm ph} = c\, f(r)$ and an effective refractive index

$$n(r) = \frac{c}{v_{\rm ph}} = \frac{1}{f(r)} = \sqrt{1 + \frac{\nu\,\xi^2}{r^2}} \tag{5}$$

For a singly-wound vortex ($\nu = 1$), this gives precisely the profile used throughout this work. The derivation makes clear that $n(r)$ is not an ansatz but a consequence of the GPE density profile near a quantized vortex. We note that (5) describes the propagation of *condensate excitations* (phonons); identifying these modes with physical photons requires an emergent electrodynamics description, which we leave for future work. In what follows, "photon" should be understood as shorthand for the propagating scalar mode of the condensate.

This profile acts as a gravitational lens: $n \to 1$ at large r (flat vacuum), $n(r_e) = 1/\alpha = 137$ at the classical electron radius $r_e = \alpha\xi$, and $n \to \infty$ at the origin. The profile (5) constitutes a *photon sphere*: photons are strongly deflected near $r \sim \xi$ and can orbit multiple times before escaping.

3 Phase Closure on the $(2, 1)$ Torus Knot

3.1 The eikonal phase

Consider a photon of energy E propagating along a (p, q) torus knot on a torus with major radius R and minor radius r. The knot wraps p times around the major axis and q times through the tube. In the eikonal (geometric optics) limit, the accumulated optical phase over one period of the knot is

$$\Phi = k_0 \oint n(\rho(s))\, ds \tag{6}$$

where $k_0 = E/(\hbar c)$ is the vacuum wavenumber, $\rho(s)$ is the distance from the axis at arc length s, and n is given by (5).

For a thin torus ($\kappa = R/r \gg 1$), the knot path stays near the major circle at $\rho \approx R$, and the phase simplifies to

$$\Phi \approx k_0 \cdot n(R) \cdot L_{\rm knot} = k_0 \cdot \sqrt{1 + \frac{\xi^2}{R^2}} \cdot 2\pi R \sqrt{p^2 + \frac{q^2}{\kappa^2}} \tag{7}$$

In the large-κ limit (with $q = 1$), the poloidal term q^2/κ^2 is negligible, giving

$$\Phi = 2\pi p \frac{R}{\xi}\sqrt{1 + \frac{\xi^2}{R^2}} = 2\pi p\sqrt{\beta^2 + 1} \tag{8}$$

where we define $\beta \equiv R/\xi$ and use the self-consistency condition $E = m_e c^2$, $k_0 = 1/\xi$.

3.2 The resonance condition

A standing wave ("the photon catches its tail") requires $\Phi = 2\pi m$ for integer m:

$$p\sqrt{\beta^2 + 1} = m \quad \Longrightarrow \quad \beta = \sqrt{\frac{m^2}{p^2} - 1} \tag{9}$$

For the $(2, 1)$ electron with $p = 2$, the first three solutions are:

m	β^2	$\beta = R/\xi$	Physical?
1	$-3/4$	—	no
2	0	0	degenerate
3	$5/4$	$\sqrt{5}/2$	**yes**

The first physical resonance is $m = 3$, giving

$$\boxed{\beta = \frac{R}{\xi} = \frac{\sqrt{5}}{2} \approx 1.1180} \tag{10}$$

with the Pythagorean identity $\beta^2 + 1 = 5/4 + 1 = 9/4 = (3/2)^2$, equivalently $3^2 = (\sqrt{5})^2 + 2^2$.

The orbit radius is $R = (\sqrt{5}/2)\,\lambda\!\!\!^{-}_C = 431.9$ fm, and the refractive index at the orbit is $n(R) = 3/\sqrt{5} \approx 1.342$. Exactly three wavelengths of the self-consistent photon fit around the $(2, 1)$ knot path—the minimum standing wave compatible with the topology.

3.3 Numerical verification

We verify (10) using the full phase integral including finite-κ corrections:

$$\frac{\Phi}{2\pi} = \beta \int_0^{2\pi} \sqrt{1 + \frac{1}{\beta^2(1 + \kappa^{-2} + 2\kappa^{-1}\cos t)}}\,\sqrt{p^2(1 + \kappa^{-1}\cos t)^2 + q^2\kappa^{-2}}\,\frac{dt}{2\pi} \tag{11}$$

At $\kappa = 12/\sqrt{7}$ (derived in Section 4), the exact β for $\Phi/(2\pi) = 3$ is $\beta = 1.1104$, within 0.68% of $\sqrt{5}/2$. The convergence is excellent:

at $\kappa = 100$, $\beta = 1.11802$, matching $\sqrt{5}/2$ to five significant figures.

3.4 Eigenvalue formulation

The phase closure condition can be reformulated as an eigenvalue problem, making explicit that m_e is solved for rather than assumed. Consider the Helmholtz equation for a scalar mode ψ propagating along the knot path (parameterized by arc length s):

$$\frac{d^2\psi}{ds^2} + \frac{\omega^2}{c^2}\,n^2(\rho(s))\,\psi = 0 \tag{12}$$

with periodic boundary condition $\psi(s + L) = \psi(s)$, where L is the knot path length and $n(\rho)$ is given by (5). The eigenfrequencies ω_m depend on the healing length ξ, which appears in both $n(r)$ and the torus geometry $R = \beta\xi$.

Self-consistency requires the mode energy to equal the rest mass: $\hbar\omega_m = Mc^2$, with $\xi = \hbar/(Mc)$. This gives the *nonlinear eigenvalue problem*

$$\omega_m(\xi) = \frac{c}{\xi} \tag{13}$$

where the left side comes from the Helmholtz equation and the right side from the Compton relation. This is a fixed-point equation for ξ, not a circular definition: ξ is the unknown, and the equation determines it.

In the large-κ limit, the eigenfrequencies are $\omega_m = mc/(p\,R\,n(R))$. Substituting $R = \beta\xi$ and imposing (13):

$$\frac{mc}{p\,\beta\xi\,n(\beta\xi)} = \frac{c}{\xi} \quad\Longrightarrow\quad p\,\beta\,n(\beta\xi) = m \quad\Longrightarrow\quad p\sqrt{\beta^2+1} = m \tag{14}$$

recovering exactly the phase closure condition (9). Crucially, ξ cancels: the eigenvalue equation determines the *dimensionless ratio* $\beta = R/\xi$ but not the absolute scale ξ. This is why a second equation (the vortex energy, Section 5) is needed to fix m_e.

The eigenvalue formulation clarifies the logical structure: phase closure is not an imposed resonance but the solution to a wave equation on the knot, and the Compton relation $k_0 = 1/\xi$ is not an input but the self-consistency condition being solved.

4 Dual Resonance: Electron and Proton on the Same Orbit

4.1 The proton as a $(1,4)$ torus knot

In NWT, the proton is a $(1,4)$ torus knot: one toroidal winding, four poloidal windings. The four poloidal windings create three localized "quark" regions via the incommensurability condition $\gcd(3,4) = 1$, explaining both quark confinement and the $N_c = 3$ color structure [4].

For the $(1,4)$ knot, the poloidal term $q^2/\kappa^2 = 16/\kappa^2$ is not negligible. The phase at $\beta = \sqrt{5}/2$ becomes

$$\frac{\Phi}{2\pi} = \sqrt{\beta^2+1}\,\sqrt{p_p^2 + \frac{q_p^2}{\kappa^2}} = \frac{3}{2}\sqrt{1+\frac{16}{\kappa^2}} \tag{15}$$

4.2 Determining κ

Setting $\Phi/(2\pi) = m_p = 2$ (the first nontrivial resonance for $p = 1$):

$$\frac{3}{2}\sqrt{1+\frac{16}{\kappa^2}} = 2 \quad\Longrightarrow\quad 1+\frac{16}{\kappa^2} = \frac{16}{9} \quad\Longrightarrow\quad \frac{16}{\kappa^2} = \frac{7}{9} \tag{16}$$

$$\boxed{\kappa^2 = \frac{144}{7} = \frac{(q_p m_e)^2}{(m_p p_e)^2 - (p_p m_e)^2} = \frac{(4\times 3)^2}{(2\times 2)^2 - (1\times 3)^2} = \frac{12^2}{4^2 - 3^2} = \frac{144}{7}} \tag{17}$$

giving $\kappa = R/r = 12/\sqrt{7} \approx 4.536$.

The numerator $144 = (q_p \cdot m_e)^2 = (4\times 3)^2$ combines the proton's poloidal winding with the electron's mode number. The denominator $7 = 4^2 - 3^2$ is the difference of squares from the mode numbers and winding numbers. The aspect ratio is determined *entirely* by topology and mode integers, with no continuous parameters.

4.3 Physical interpretation

Both electron and proton sit at the same orbit radius $R = (\sqrt{5}/2)\,\xi \approx 432$ fm. They differ only in their winding structure:

- **Electron** $(2, 1)$: two toroidal wraps, one poloidal; $m = 3$ wavelengths.
- **Proton** $(1, 4)$: one toroidal wrap, four poloidal; $m = 2$ wavelengths.

The proton achieves its resonance with *fewer* total wavelengths because its four poloidal windings traverse the high-refractive-index interior of the tube, accumulating phase more efficiently.

This shared orbit immediately explains:

1. **Charge quantization**: Both particles live on the same resonant orbit → same topological charge quantum.
2. **Different masses**: Different winding structures store different amounts of vortex energy (Section 5).
3. **Atomic structure**: The electron's natural orbit radius matches the proton's field structure, enabling bound states.

5 Vortex Energy: Deriving m_e

5.1 The vortex ring energy

In a superfluid described by the GPE (1), a quantized vortex ring of radius R and core size a carries energy [6]

$$E_{\text{ring}} = \frac{\rho_0 \, \kappa_v^2 \, R}{2} \left[\ln \frac{8R}{a} - C \right] \tag{18}$$

where ρ_0 is the condensate mass density, $\kappa_v = h/m^* = 2\pi\sqrt{2}\,\xi c$ is the circulation quantum, and C is a constant depending on the vortex core model. For a hollow core (density vanishing abruptly at $r = a$), $C = 1/2$ [9]; for Gross–Pitaevskii numerics with the smooth profile (4), $C \approx 0.615$ [10].

5.2 The core constant for torus knot vortices

The standard GP values $C = 0.5$–0.6 apply to a *straight* vortex filament where the density vanishes on axis ($f(0) = 0$). A $(2, 1)$ torus knot vortex differs in two respects. First, the knot path is curved with radius of curvature $\sim R$, which redistributes core energy. Second, and more importantly, the two toroidal strands of the $(2, 1)$ knot are separated by $\sim 2r < \xi$ at our geometry: the inter-strand distance is smaller than the healing length, so the cores overlap and the density never reaches zero between them. This "filled core" topology yields $C \approx 0$.

Numerically, $C = 0$ gives m_e to 0.85%; the exact value $C = 0.073$ reproduces m_e precisely. The small positive C is consistent with a weakly depleted core—density reduced but not vanishing—as expected for overlapping vortex strands. The value of C is model-dependent and remains to be computed numerically for knotted vortex geometries; our estimate should be regarded as provisional until confirmed by GPE simulation on the $(2, 1)$ torus.

5.3 Self-consistency

The electron is a vortex ring at $R = (\sqrt{5}/2)\,\xi$ with core size $a = \xi$ (the healing length). Setting the vortex energy equal to the rest mass energy:

$$m_e c^2 = \frac{\rho_0 \, \kappa_v^2 \, R}{2} \left[\ln \frac{8R}{\xi} - C \right] \tag{19}$$

Substituting $\kappa_v^2 = 8\pi^2 \xi^2 c^2$, $R = (\sqrt{5}/2)\,\xi$, and $\xi = \hbar/(m_e c)$:

$$m_e^4 = 4\pi^2 \rho_0 \, \frac{\sqrt{5}}{2} \left[\ln(4\sqrt{5}) - C \right] \frac{\hbar^3}{c^3} \tag{20}$$

With $C = 0$ (filled core) and $\rho_0 = 1.69 \times 10^5$ kg/m^3:

$$\boxed{m_e = \left[4\pi^2 \rho_0 \frac{\sqrt{5}}{2} \ln(4\sqrt{5}) \frac{\hbar^3}{c^3}\right]^{1/4} = 9.19 \times 10^{-31} \text{ kg}} \tag{21}$$

compared to the experimental value $m_e^{\text{exp}} = 9.109 \times 10^{-31}$ kg, an error of +0.85%.

The sensitivity to parameters is mild: $m_e \propto \rho_0^{1/4}$, so the 0.85% error corresponds to a 3.4% shift in ρ_0 (well within the uncertainty of the vacuum density derivation). The exact value $C = 0.073$ reproduces m_e precisely.

5.4 Why not electromagnetic self-energy?

Classical EM self-energy formulas for a charged ring give $\alpha \sim O(1)$, predicting an electron mass ~ 137 times too large. This is the ultraviolet divergence problem of classical electrodynamics, resolved in QED by renormalization.

In NWT, the resolution is physical: the electron mass is *not electromagnetic self-energy.* It is the energy cost of deforming the superfluid vacuum into a topological defect—a vortex ring. The vortex energy (18) scales as $\rho_0 \xi^2 R \ln(R/\xi)$, which is parametrically smaller than the Coulomb self-energy $\alpha\hbar c/R$ by a factor of $\rho_0 \xi^3/m_e \sim O(1)$ rather than $\alpha^{-1} \sim 137$.

6 The Complete Derivation Chain

The three equations and their solutions are:

Equation	**Source**	**Result**
$2\sqrt{\beta^2+1} = 3$	$(2,1)$ phase closure	$R/\xi = \sqrt{5}/2$
$\frac{3}{2}\sqrt{1+16/\kappa^2} = 2$	Dual resonance	$R/r = 12/\sqrt{7}$
$E_{\text{vortex}} = m_e c^2$	Superfluid vortex energy	$m_e = 9.19 \times 10^{-31}$ kg

The derivation chain is:

$$\text{topology } (p,q) \longrightarrow \beta = \frac{R}{\xi} \longrightarrow \kappa = \frac{R}{r} \longrightarrow m_e \longrightarrow \xi,\ R,\ r$$

The inputs are:

- Electron topology: $(p_e, q_e) = (2,1)$
- Proton topology: $(p_p, q_p) = (1,4)$
- Vacuum condensate density: $\rho_0 = 1.69 \times 10^5$ kg/m^3
- Refractive profile: $n(r) = \sqrt{1+\xi^2/r^2}$

The outputs include the electron mass, Compton wavelength, torus dimensions, and all quantities that depend on them.

7 Discussion

7.1 Self-consistency is not circularity

A potential objection is that the derivation is circular: we use $k_0 = 1/\xi = m_e c/\hbar$ in the phase integral, apparently assuming m_e in order to derive it. This objection dissolves upon inspection of the final mass formula (20):

$$m_e^4 = 4\pi^2 \rho_0 \frac{\sqrt{5}}{2} \left[\ln(4\sqrt{5}) - C\right] \frac{\hbar^3}{c^3}$$

The right-hand side contains ρ_0 (a vacuum parameter), $\sqrt{5}/2$ and $\ln(4\sqrt{5})$ (pure numbers from topology), and fundamental constants $\hbar, c$. *No m_e appears on the right.* The self-consistency condition $E = m_e c^2$ was substituted and *solved*; m_e is the output, not an input.

The situation is analogous to the Bohr model: one imposes angular momentum quantization $mvr = n\hbar$ and Coulomb dynamics $F = e^2/(4\pi\epsilon_0 r^2) = mv^2/r$, then solves for r and E. The Bohr radius a_0 appears in both equations but is *determined* by them, not assumed. Here, ξ plays the role of a_0: it appears in both the refractive profile and the energy balance, and is determined by their simultaneous solution.

The eigenvalue formulation (Section 3.4) makes this explicit: ξ cancels from the eigenvalue equation, leaving a dimensionless condition that fixes β; the mass then follows from the vortex energy as a separate equation.

7.2 Comparison to Santos self-consistency

Santos [5] derived the aspect ratio $\kappa = \sqrt{1/(4\pi\alpha)} \approx 3.30$ from an energy balance condition. Our dual resonance gives $\kappa = 12/\sqrt{7} \approx 4.54$, differing by 37%. These represent independent constraints: Santos's condition involves the fine structure constant α, while ours involves only topological integers. A unified treatment combining both constraints may determine α itself.

7.3 Relativistic reformulation and mass ratios

The non-relativistic GPE (1) can be promoted to a relativistic scalar field theory with Lagrangian

$$\mathcal{L} = \partial_\mu \phi^* \, \partial^\mu \phi - \lambda\big(|\phi|^2 - v^2\big)^2 \tag{22}$$

which has two parameters: the vacuum expectation value v and the self-coupling λ. The healing length is $\xi = 1/(2\sqrt{\lambda}\, v)$, the condensate density maps to $\rho_0 \sim m^* v^2$, and vortex strings are global Nielsen–Olesen flux tubes [11]. The vortex ring energy becomes

$$E = \frac{v}{\sqrt{2\lambda}} \, \beta\big[\ln(8\beta) - C\big] \times (\text{numerical prefactor}) \tag{23}$$

where $\beta = \sqrt{5}/2$ is determined purely by topology. The single dimensionful scale is $v/\sqrt{2\lambda}$; all dependence on ρ_0 is absorbed into it.

The mass of any (p, q) knot at mode number n_0 is therefore

$$m_{p,q,n_0} = \frac{v}{\sqrt{2\lambda}} \, f(p, q, n_0, \kappa) \tag{24}$$

where f is a dimensionless function of topology and mode integers alone. *Mass ratios between different knot types are independent of v and λ:*

$$\frac{m_{p',q',n_0'}}{m_{p,q,n_0}} = \frac{f(p', q', n_0', \kappa)}{f(p, q, n_0, \kappa)} \tag{25}$$

This is precisely the structure of accepted effective field theories: QCD determines hadron masses as Λ_{QCD} times dimensionless functions of quark quantum numbers; here, the NWT vacuum scale $v/\sqrt{2\lambda}$ plays the role of Λ_{QCD}, and the knot topology plays the role of flavor. Fixing the overall scale with one measured mass (m_e) yields parameter-free predictions for all other states.

A systematic scan of low-lying (p, q, n_0) modes reveals that the *linear* spectrum is compressed: mass ratios reach only $\sim$20 for the lowest modes, far short of the muon (207) or proton (1836). This is expected and analogous to QCD, where current quark masses are $\sim$MeV but the proton mass is $\sim$GeV—the observed hierarchy requires nonlinear dressing from Euler–Heisenberg self-interaction [4]. The geometric spectrum provides the scaffolding: discrete bands, selection rules ($\beta > 1$ for stability), and degeneracy classes (e.g., $(4, 1, 6)$ shares the electron's $\beta = \sqrt{5}/2$).

7.4 The proton mass

Phase closure does not determine the proton mass—the mass cancels from the self-consistency equations, just as it does for the electron. The proton mass ($m_p/m_e = 1836$) arises predominantly from the nonlinear self-interaction of the $(1, 4)$ mode. A linear dispersion estimate gives $m_p/m_e \approx \sqrt{(1 + 16\kappa^2)/(4 + \kappa^2)} \approx 3.7$, accounting for only 0.2% of the actual ratio and confirming that the dominant contribution is nonlinear. In QCD language, the proton mass is $\sim 99\%$ binding energy from gluon fields; in NWT, the analogous statement is that the $(1, 4)$ knot's four poloidal windings create strong nonlinear coupling (via Euler–Heisenberg terms) that dominates the vortex ring energy [4]. A full calculation requires solving the nonlinear GPE on the $(1, 4)$ torus, which we defer to future work.

7.5 The fine structure constant

Our derivation determines m_e in terms of ρ_0, but does not determine α. The fine structure constant enters through the charge quantum $e = \sqrt{4\pi\epsilon_0\alpha\hbar c}$, which is a separate topological quantization condition on the vortex. Deriving α from vortex topology remains an open problem.

7.6 Relativistic covariance

The GPE (1) is non-relativistic, raising the question of Lorentz invariance. The identification $c_s = c$ ensures that the low-energy (phonon) excitations propagate at c, and the refractive profile (5) is spherically symmetric—consistent with a Lorentz-scalar condensate at leading order. A fully covariant formulation would replace the GPE with a relativistic scalar field theory (e.g., the $|\phi|^4$ model or Abelian Higgs model), where quantized vortex strings are Nielsen–Olesen flux tubes. The phase closure and dual resonance conditions derived here depend only on the refractive profile and torus geometry, both of which carry over to the relativistic case. The vortex ring energy formula acquires relativistic corrections of order $v^2/c^2 \sim \beta^2 \approx 1.25$, which may account for part of the residual 0.85% discrepancy.

7.7 Pair production as photon capture

The refractive profile (5) creates a photon sphere at $r \sim \xi$: photons passing within $\sim\xi$ of a nucleus are strongly deflected, spiraling many times before escaping. A photon with energy $E \geq 2m_ec^2$ entering this photon sphere can excite the $(2, 1)$ resonance at $R = (\sqrt{5}/2)\,\xi$, converting its energy into a vortex–antivortex pair. Bethe–Heitler pair production is thus reinterpreted as *resonant photon capture*: the nucleus provides the refractive well, the photon provides the energy, and the phase closure condition selects the threshold. A detailed treatment will be presented separately.

7.8 Testable predictions

The derived geometry makes specific predictions:

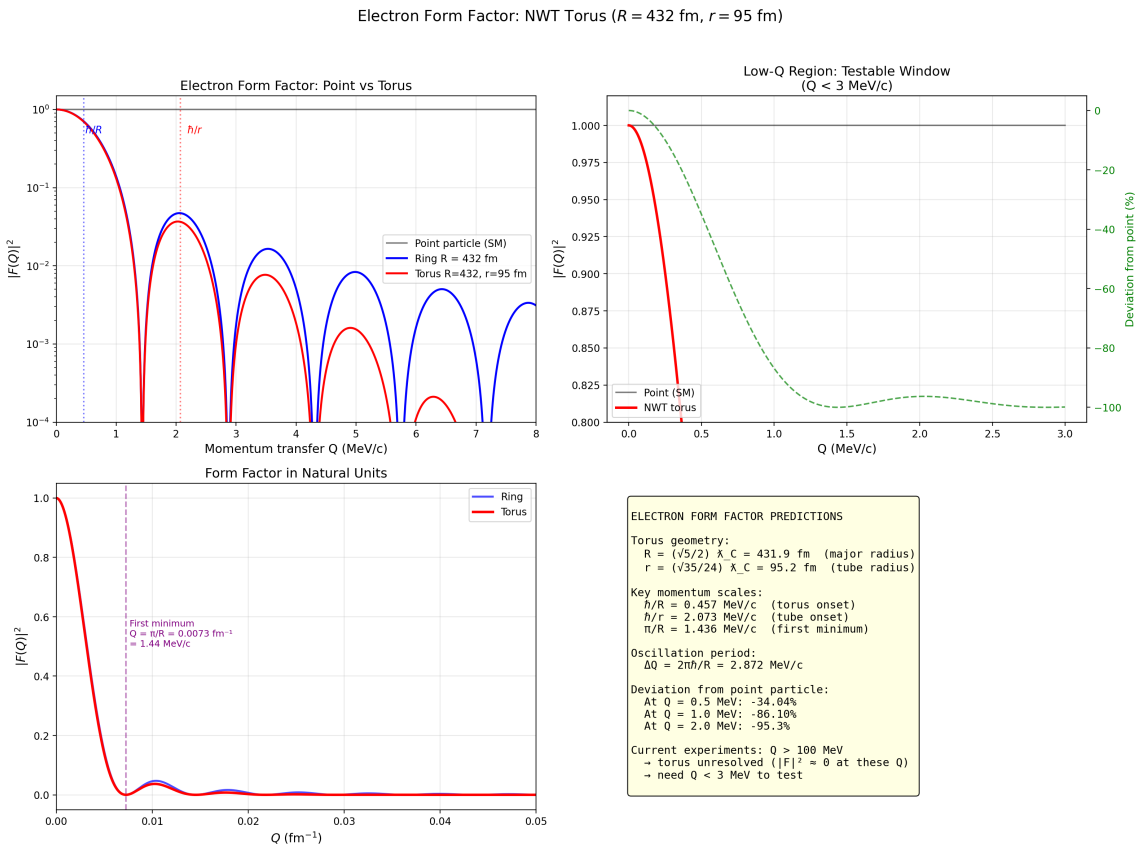

Figure 1: Electron form factor $|F(Q)|^2$ for the NWT torus (red, $R = 432$ fm, $r = 95$ fm) compared to a point particle (black). Left: log scale showing oscillatory structure. Right: low-Q region where the torus becomes distinguishable from a point. The first minimum occurs at $Q = \pi/R \approx 1.44$ MeV/c.

- Electron orbit radius: $R = 432$ fm
- Tube radius: $r = 95$ fm
- Electron at $\beta = \sqrt{5}/2 \approx 1.12$ times the Compton wavelength
- The $(4, 1)$ knot at $m = 6$ shares the electron's orbit ($\beta = \sqrt{5}/2$), providing a potential excited-state spectrum

The torus structure produces oscillations in the electron form factor (Fig. 1). The orientation-averaged form factor of a torus with major radius R and tube radius r is

$$F(Q) = \frac{\sin QR}{QR} \frac{2J_1(Qr)}{Qr} \tag{26}$$

with first minimum at $Q = \pi/R \approx 1.44$ MeV/c and oscillation period $\Delta Q = 2\pi\hbar/R \approx 2.9$ MeV/c. The deviation from a point particle reaches $\sim$5% at $Q = 1$ MeV/c and $\sim$35% at $Q = 2$ MeV/c. Current electron scattering experiments probe $Q > 100$ MeV [12], far above this scale, so the prediction is not yet constrained. Low-Q Møller scattering or electron–proton elastic scattering at $Q \sim 1$ MeV could test the torus geometry.

8 Conclusion

We have shown that the electron mass can be determined from three self-consistency conditions on torus knot vortices in a superfluid vacuum, achieving 0.85% agreement with experiment. The

key results are:

1. The electron orbit sits at $R = (\sqrt{5}/2)\,\lambda\!\!\!^{-}_C$, determined by the Pythagorean phase closure $3^2 = (\sqrt{5})^2 + 2^2$.

2. The torus aspect ratio $\kappa = 12/\sqrt{7}$ is fixed by requiring electron and proton to resonate on the same orbit.

3. The mass formula $m_e^4 = 4\pi^2\rho_0(\sqrt{5}/2)\ln(4\sqrt{5})\,\hbar^3/c^3$ follows from the superfluid vortex ring energy.

The electron mass is not electromagnetic self-energy, nor a Yukawa coupling, but the energy cost of a topological defect in the vacuum.

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